

Artificial Intelligence & Machine Learning

Task 5: Ensemble Learning & Model Optimization

Dataset: Breast Cancer Dataset (Scikit-learn)

Models: Logistic Regression, Random Forest, Gradient Boosting

1. Introduction

This project is about ensemble learning and model optimization. In this task, I trained different classification models and compared their performance. I also used GridSearchCV and cross-validation to check and improve the Random Forest model.

2. Objective

The main objective of this project was to understand ensemble learning, compare different models, tune model parameters and evaluate the models using different performance metrics.

3. Tools Used

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook and Joblib.

4. Models Used

Logistic Regression was used as the baseline model. Random Forest and Gradient Boosting were used as ensemble learning models.

5. Model Results

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	98.25%	98.61%	98.61%	98.61%
Random Forest	95.61%	95.89%	97.22%	96.55%
Gradient Boosting	95.61%	94.67%	98.61%	96.60%

From the initial model comparison, Logistic Regression gave the highest test accuracy (98.25%) and F1 Score (98.61%). Random Forest and Gradient Boosting both achieved 95.61% accuracy.

6. Hyperparameter Tuning

Random Forest was optimized using GridSearchCV. The best parameters were **max_depth = 5** and **n_estimators = 100**. The best CV score was **96.33%**.

7. Cross-Validation

The five cross-validation scores were 0.96703297, 0.98901099, 0.92307692, 0.92307692 and 0.94505495. The mean CV accuracy was 94.95% and the standard deviation was 2.56%.

8. Feature Importance

The feature importance graph shows the top features used by the optimized Random Forest. Worst area was the most important feature among the top ten features shown.

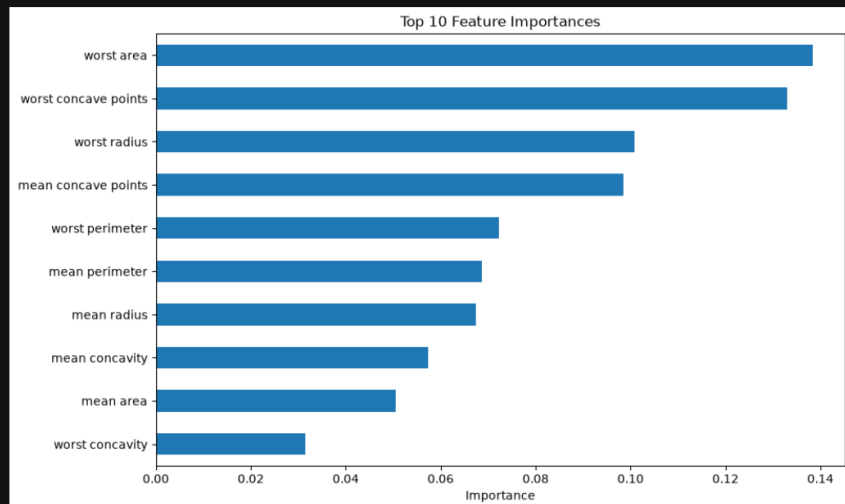


Figure 1: Top 10 Feature Importances

9. Confusion Matrix — Optimized Random Forest

The optimized Random Forest confusion matrix contains 39 true negatives, 3 false positives, 2 false negatives and 70 true positives. From this matrix, the optimized Random Forest achieved **97.37% accuracy** on the evaluated test set.

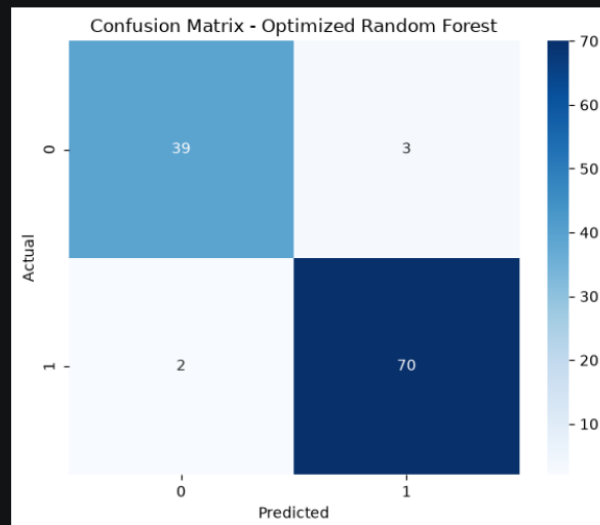


Figure 2: Confusion Matrix - Optimized Random Forest

The 95.61% Random Forest accuracy shown in the model-comparison table is the result of the initial Random Forest model. The 97.37% accuracy calculated from the confusion matrix is for the optimized Random Forest after hyperparameter tuning. Therefore, these two values refer to different versions of the Random Forest model.

10. Conclusion

In this task, I learned how ensemble models can be trained and compared. I also learned how to use GridSearchCV and cross-validation for model optimization. Logistic Regression performed best in the

initial comparison, while Random Forest was further optimized and analyzed using feature importance and a confusion matrix.